Formalising a Mathematician's Working Notes: Lean Proofs from the Grothendieck Archive

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Abstract. The University of Montpellier holds the mathematical working notes of Alexandre Grothendieck: 178 openly accessible folders, 16,074 digitised pages that its inventory dates from 1953 to 1991. An edition in progress has transcribed 9,414 of these pages with a large language model and restated 164 folders in current notation. Statements drawn from those restatements were then formalised in Lean 4 against a pinned revision of Mathlib, in 74 files (72 folders and two Bourbaki drafts) holding 2,146 top-level theorems that depend only on Lean's three standard axioms; 74 cards give each formalised statement its page and a verdict. We describe the pipeline from page to theorem, the rule that ties each theorem to a page without attributing to Grothendieck what the edition supplied, and a triage of 203 open statements by reachability. The formalisation found false glosses in the machine readings, claims on the pages that it refutes, slips that it corrects, and statements already in print. For one table of the notes it proves the trivialising exponents for every n in a homogeneous model, modulo a dictionary it does not formalise; the Picard groups involved were already in print for n even. We set out its relation to seven conjectures that appear in the notes, report how Mathlib received the general lemmas proposed to it, and state the limits of human checking: a proof shows that a reading holds together, but not that the reading is what the page says.

Keywords: Lean · Mathlib · autoformalisation · manuscripts · language models · Grothendieck

1 Introduction

Alexandre Grothendieck gave his mathematical papers to his former student Jean Malgoire in 1990 and 1995. They were deposited at the University of Montpellier in 2010 and given to it in 2012. The university has catalogued the fonds and placed 178 folders, 16,074 pages by the inventory's count, in open access as facsimiles [45]. They are working notes: drafts in ink and pencil, restarted, struck and annotated, in a notation of his own. Mathematicians have edited twenty folders by hand, among them *Les Dérivateurs* [34] and the first volume of *À la poursuite des champs* [36]; the rest had not been transcribed.

Since August 2026 an edition has read the remaining folders with a large language model as first transcriber [28,27]. At the commit this paper describes[1], 638 passes had read 9,414 pages in 164 folders into a restricted LaTeX with a critical apparatus, and every one of the 164 folders had a *modernised reading*: a restatement in current notation, in French, derived from the transcription and never from the hand. A companion paper for the *Journal of the TEI* [28] covers the encoding of the transcriptions and the checking of the readings; this one covers the formalisation of the statements. The rights in the manuscripts belong to Grothendieck's heirs. Following the reply of an institution consulted in September 2026, the reading views of the transcriptions are withheld from the public site until the heirs authorise them; the statements, the Lean files and the written-out proofs remain public, and each cites the page of Montpellier's facsimile.

A modernised reading claims that a stretch of manuscript says a particular piece of mathematics; a model makes the claim and no mathematician has checked it. Such a reading usually fails in one of four ways: an inverted variance, a hypothesis the page carries and the reading drops, an inclusion turned into an equality, or a modern name applied to something it does not fit. A proof assistant refuses all four. Lean cannot say whether the reading is faithful to the page, because a misreading can be formalised as exactly as a correct reading. It can say whether what the reading states holds, under which hypotheses, and with which counterexample when it does not.

The contributions are: (i) a pipeline from facsimile to Lean theorem, with a rule of attribution linking each theorem to a page (Sections 2 and 3); (ii) a triage of the open statements by what Mathlib can reach (Section 4); (iii) an account of what the formalisation revealed about the readings and the pages, and of what it can do with the conjectures in the notes (Sections 5 and 6); (iv) an account of what the edition contributes to the Tau Ceti roadmaps (Section 7); (v) an account of human checking and of costs (Sections 8 and 9).

2 Corpus and Pipeline

Figure 1 shows the chain. The transcription is written twenty pages at a time in a LaTeX subset with nine apparatus macros (illegible, uncertain, supplied, added, struck, note and the like); two scripts refuse any construct outside the subset. The modernised reading of a folder is written in one pass over all its batches, with footnotes that carry what the apparatus carried. A further register, the *findings* (`src/content/findings.ts`), lists statements of the folders that may not stand in the literature, each written as a claim about the literature with what was searched and what would settle it. It holds 463 rows from 155 folders: 216 unsearched, 165 candidates, 73 matched, 8 refuted and 1 confirmed, a status

[1] Commit `81dcfaf` of the repository [27], 24 October 2026. Every count in the paper is taken at that commit, most of them by `docs/cicm/figures.mjs`, which writes them into the source of the paper.

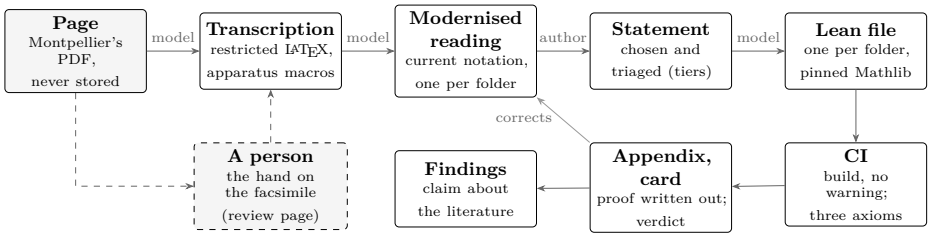


Fig. 1. From the page to the theorem. Every arrow marked *model* is written by a language model under written rules; the author chooses the statements among those the model proposes, and the dashed loop is the only place a person compares the text with the page. A failed proof or a counterexample goes back to the modernised reading (grey arrow), never to the transcription, which only a reading of the hand may change.

only a person may set. The language models used at each step, and their roles, are listed in the declaration at the end of the paper.

Statements are taken from the readings and the findings and formalised in `lean/Grothendieck/`, one file per folder (`Folder19.lean`, `Folder161_2.lean`), plus two files, `Bourbaki288.lean` and `Bourbaki457.lean`, for drafts of the Bourbaki archive [1]. Each file opens with a docstring that names the folder, the pages and what is and is not proved, and keeps its own namespace. There are 74 files (72 folders and the two Bourbaki drafts), with 41,553 lines and 2,146 top-level `theorem` declarations, counted by `grep -c '^theorem'` over these files, the Bourbaki drafts included. This count measures size, not statements: it includes auxiliary results and the edition's own lemmas and counterexamples, and it leaves out 307 top-level `lemma` declarations, 9 private theorems, and the 85 declarations of the general lemmas that the proofs share, kept apart in 26 files under `lean/Grothendieck/Upstream/` (Section 7). Table 1 sets it beside the statements that the appendix attributes. The first file was committed on 26 September 2026, and 206 commits had touched `lean/` by the commit described here.

Mathlib is pinned to a revision of its `master` branch, not to a release, with the toolchain `leanprover/lean4:v4.35.0-rc4`, so that the proofs are checked against the library that Tau Ceti builds on (Section 7).[2] Moving the pin is a deliberate step: a deprecation fails the build, and the files are rebuilt one at a time because each process importing Mathlib needs about 2 GB. A continuous integration workflow (`.github/workflows/lean.yml`) builds the library with warnings as errors, so that a `sorry` fails it, and then runs the axiom audit of `lean-action`. The audit covers the declarations under the project's root namespace, `Grothendieck`, and fails if one of them depends on an axiom other than `propext`, `Classical.choice` and `Quot.sound`; `sorry` and `native_decide` count as such axioms. The general lemmas of `Upstream/` are declared in Mathlib's namespaces, so they are audited only through the theorems that use them. The

[2] Mathlib commit `715a03a90adf491de40fbec8b456c179a623435f`.

Table 1. What the formalisation contains. The Lean counts are of top-level declarations; the appendix counts are of its boxed statements, by the label that the attribution rule gives them.

Lean library		Appendix	
Files: folders, Bourbaki drafts	72, 2	Theorems, « Grothendieck, page N »	49
`theorem` declarations	2,146	Theorems, « d'après la page N »	21
`lemma` declarations	307	Theorems, « énoncé de l'édition »	23
`Upstream/`: files, declarations	26, 85	Counterexamples	25
Cards	74	Other statements of the readings	131
		Lemmas	74

independent kernel re-check, `leanchecker`, is not run in CI, because it stopped the runner. No file declares an axiom or uses `native_decide`.

3 Attribution and the Link from Page to Theorem

A formalisation of working notes must keep apart three things: what the page states, what the reading states, and what the formaliser had to add. The edition's rule is that a statement is labelled « Grothendieck, page N » only if that page states it. A result the edition established to make a statement provable (a lemma, a weaker hypothesis that suffices, a counterexample) gets a neutral label and never his name. The same rule governs the pull requests to Mathlib: Grothendieck is named only for statements genuinely his, in one line of the description, never in a docstring.

Every proof has three human-readable forms. The Lean file is the record. A card in `src/content/proofs.ts`, rendered on the site, gives the folder, the batch and page, the reading's name for the statement, the statement, what the formalisation found, and a verdict from a closed list. A card whose statements fare differently carries one verdict for each, as 6 do. Of the 74 cards, 12 carry *holds*, 16 *holds, more generally*, 36 *partly proved*, 3 *the reading dropped a hypothesis*, 4 *holds; the gloss was false*, 8 *false on the page* and 1 *conjecture of the page refuted*. The third form is an appendix in French [26], written for a reader who is a mathematician but not a Lean user: for each folder, the leaf in context, the statement as the reading gives it, a proof written out with its intuition, remarks, and what is not proved. Its 298 pages are divided into parts that follow the themes of the fonds in chronological order (Figure 2), and the general lemmas are cited by their file in the repository instead of being proved again. The appendix labels each boxed statement by the rule above (Table 1). A theorem is labelled « Grothendieck, page N » when the page states it, « d'après la page N » when the proof establishes more than the page, and « énoncé de l'édition » when the edition found it; a counterexample refutes a claim of the page or of the reading; the other statements of the readings, proved in part or in a model, keep the name *énoncé*. Where a formalisation bears on a finding, the finding's row records it, and a counterexample moves the row to *refuted* with what still stands.

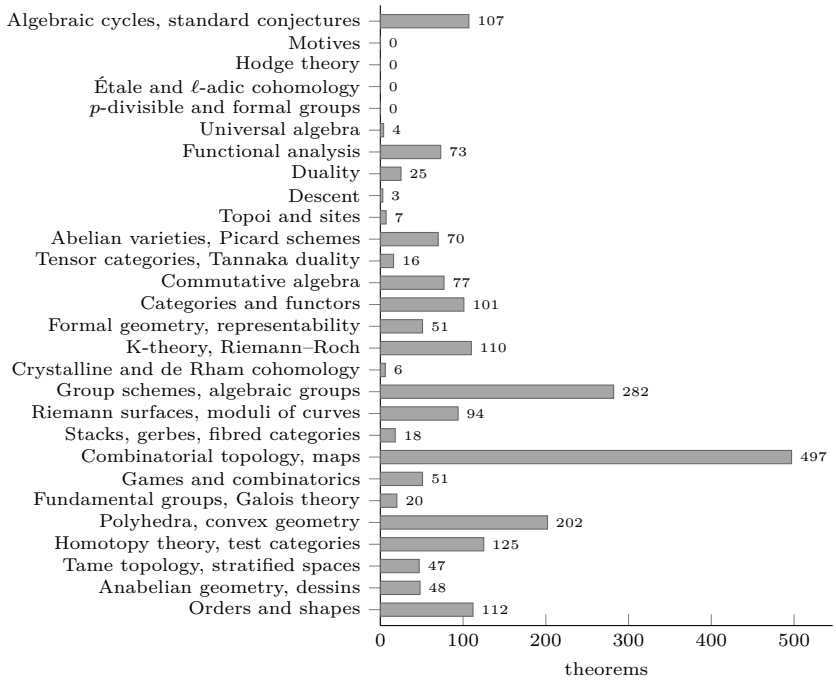


Fig. 2. Lean theorems per theme, the themes being the 28 parts of the written-out appendix in their order (2,146 of the 2,146 theorems; every file belongs to a part). The 4 parts at zero hold only statements left unproved.

4 Triage: What Mathlib Reaches

The first eighteen folders were chosen by hand (issue #26) as those whose readings would repay a proof. In October 2026 the 203 findings that were still open and had no proof, in 111 folders, were sorted by reachability (issue #54). Tier 1, reachable now with Mathlib as it stands, held 16; tier 2, a project of a few hundred lines in which part of the statement is reachable, 39; tier 3, out of reach, 147; and one finding was ill-posed as read and set aside to be disproved. A tier says nothing about whether a statement is new.

Most of the corpus is in tier 3 because its statements live in theories that Mathlib does not yet have, or has only in foundational form: schemes beyond their definition and first properties, étale and ℓ-adic cohomology, crystalline cohomology, topoi and their morphisms, stacks and gerbes, derivators, motives, and Hodge structures on cohomology. Figure 2 shows the effect. Four parts of the appendix (motives, Hodge theory, étale and ℓ-adic cohomology, p-divisible groups) contain no proved statement at all, and descent only the field case of a lemma (folder 128), while combinatorial topology, polyhedra, group schemes taken in coordinates, and homotopy theory through test categories carry most of

the theorems. The distribution follows Mathlib's coverage; Grothendieck's own centre of gravity in these folders lies in the parts that are empty.

What can be reached is often a special case or a model. A group scheme is treated on R-points in coordinates over an arbitrary ring (folder 72), a statement about schemes is proved for $\operatorname{Spec} A$ with A Noetherian (folder 42), and a remark about connections is proved on the trivial bundle only (folder 93). Of the 74 cards, 36 say *partly proved*, and each says what is not. Several others prove a statement only in a model, or in the special case that Mathlib reaches, and say so. We count these as partial too, since a proof in a model checks that the axioms of the page are consistent and that a computation is right without proving the general statement. The issues that track the remaining rows (#59, #72) give each an estimated size and a value, and drop those whose only reachable part would be a special case of the edition's choosing.

5 What Formalisation Revealed

False glosses in the readings. Most often the page is right and the reading, where it goes beyond the page, is wrong. In folder 158 the Proposition holds as the leaf states it, finiteness being used exactly where the page says « comme E_0 est fini », but the modern gloss called an element a left zero and the Rees kernel a point; in the monoid $\{1, a, b\}$ with $xa = a$ and $xb = b$ both a and b qualify, the kernel is $\{a, b\}$, and the element is a right zero, not unique. In folder 63 the reading took a sentence of page 99 that begins in mid-phrase to say that four components of a modular curve share their tangent at a point over $\mathbb{F}_4$; the four tangents are pairwise distinct. In folder 114 the reading dropped the page's « satisfying Loc 4) » from the hypotheses of a statement on fundamental localisers, without which the inclusion $W \subseteq W_0$ fails. In folder 74 it left out an axiom written in the margin, whose end is illegible: under a reconstruction of the axiom the reduction of pages 92–103 holds, and without it the claim of page 97 fails for a Latin square of order 2. In folder 16 the reading's universal ring $\Phi_0 = \mathbb{Z}[L_0, \Lambda_0]$ does not exist over $\mathbb{Z}$ for $n \geq 2$, since ξ_0^n is divisible by $n!$; over $\mathbb{Q}$ the obstruction disappears. In folder 156-3 a footnote's form of the betweenness relation is false when the extremities are incomparable, as the product order on $\mathbb{N}^2$ shows; the lemma of the page never meets that case. Each reading was corrected. A related error was found without Lean: the summary of folder 114 had presented as true Proposition 4 of page 350 of *À la poursuite des champs*, which the literature refutes by the category of cubes [11,35].

Errors on the page. We call an error on the page a claim written on the leaf, not added by the reading, that Lean refutes or corrects. The 8 cards marked *false on the page* each refute such a claim by a counterexample in Lean. In folder 19 a comonadicity theorem gives as necessary and sufficient a condition that is only sufficient: the inclusion of the pairs of sets (X, Y) with $Y \neq \emptyset \Rightarrow X \neq \emptyset$ is coreflective, hence comonadic, yet fails the condition. The reading had repeated the claim and identified the condition, wrongly, with Beck's; on page 7 the proviso

« ou I fini » is false as well. In folder 80 a theorem on crucial arrangements of pseudolines fails for six straight lines, the sides of a triangle and one further line through each vertex. In folder 90 a parenthesis on flat affine groups over a Dedekind ring fails without finite type, for the algebra $\{f \in K[T] : f(0) \in R\}$ over a discrete valuation ring R. In folder 63 the splitting asserted on page 99 is not stable under the Hesse involution, which acts as a transvection. In folder 118 the remark of page 35 that a condition is « visiblement stable par sommes » fails for the sum of two points. In folder 150 the equivalence of page 19 needs the strictness condition circled at the foot of page 16. In folder 158 the criterion of page 52 fails, as written, for the discrete category on two objects. In folder 160 a corollary that leaves a sign undetermined fails for p and n odd: over $\mathbb{F}_3$ with $n = 1$ there are two trivialisations to count, but no square root of $a_0 = -1$; with the sign $(-1)^n$ it holds. One card refutes a conjecture: in folder 156-4 the saturated parts that page 68 presumes to form a Boolean algebra form a non-distributive hexagon in a seven-element model. Four further errors are slips rather than false claims: an index and a label carried to the wrong place (folder 86), a head line that contradicts the table below it (folder 152), a translation that is wrong in characteristic 2 (folder 70), and a normalisation $ad - bc = 1$ added on page 121 of folder 93, which Lean refutes, since the determinant of the homography is a unit that need not be a square ($s = 2t$ over $\mathbb{F}_5$). The edition records all of these as errors in notes he wrote for himself.

Statements already in print. Formalisation often sends the edition back to the literature. In folder 16 the statement that the Lefschetz-type conjecture follows from $D(X)$ and the existence of *some* algebraic isomorphism $H^{2n-i}(X) \to H^i(X)$ holds by Cayley–Hamilton in Lean. The register had listed it as possibly unpublished; the row is now *matched* to Kleiman's Theorem 2.9 of 1968 [31], proved by the same argument, and to Theorem 4-1 of his 1994 survey [32].

In the other direction, a page of the notes can bear on a published result. On page 35 of folder 118 Grothendieck asks whether stability of $\mathcal{W}_A$ under finite products makes A aspherical, and answers no with a groupoid, for which $\mathcal{W}_A$ is the isomorphisms. Set beside Cisinski's Proposition 4.3.2 [11], the example shows that condition (g) of that proposition does not imply condition (c) for $A = BG$. Lean checks this in a 1-truncated model, which takes as the definition of a weak equivalence of groupoids the usual criterion: a bijection on connected components and an isomorphism on automorphism groups. In a public answer on MathOverflow, Cisinski confirmed that (g) $\Rightarrow$ (c) needs a further assumption [12].

A question of the notes, settled in a model. Page 70 of folder 146 tabulates, for the moduli spaces $M_{0,n}$, the trivialising exponents of the line bundles L_i and $L = L_1 \otimes \cdots \otimes L_n$ under the symmetric group $\mathfrak{S}_n$, with question marks against several entries. For $n = 4$ Lean checks the cocycle computation behind the table: on $U = \mathbb{P}^1 \setminus \{0, 1, \infty\}$ the six anharmonic substitutions act through $\mathfrak{S}_3$, the derivative at the moving point is a 1-cocycle whose sixth power is the coboundary of $\lambda^4(1-\lambda)^4$, and no smaller power is the coboundary of any nowhere-zero func-

tion, regular or not; for the cocycle $\operatorname{sgn} g \cdot g'(\lambda)$ of L the exponent is the page's 3. The identification of the equivariant Picard group with $H^1(\mathfrak{S}_3, \mathcal{O}(U)^*)$, the reduction from $\mathfrak{S}_4$ to $\mathfrak{S}_3$ and the formula for the action on L come from the reading and are not formalised. For $n = 5$, where the page writes « 12 ? » for both, Lean proves at the level of cocycles that L has order 4 and L_i order 12: the page's 12 is right for L_i and not minimal for L. For $n = 6$ Lean proves the orders 5 and 20, against the page's « 20 ? » and « 60 ? »; the trivialising functions come from exact linear algebra on the nine units of $M_{0,6}$, the lower bounds from fixed points where the factor is a primitive fifth root of unity or i. For every n, Lean then proves the general formula in a homogeneous model, over a field with primitive roots of unity of orders $n-1$ and $n-2$: L has order $n-1$ and L_i order $(n-1)(n-2)$. In that model a point of $\mathbb{P}^1$ is a vector of K^2, and a trivialisation of $L_i^{\otimes m}$ is a nowhere-zero function on configurations, not required to be regular, invariant under relabelling, of weight t^{2m} in the i-th vector and $\det(g)^m$ under $\mathrm{GL}_2(K)$. The trivialisations are products of brackets $[v_i\, v_j]$, a squared Vandermonde product for L; the lower bounds come from the configurations $\{0\} \cup \mu_{n-1}$ and $\{0, \infty\} \cup \mu_{n-2}$, rotated by $z \mapsto \omega z$. The dictionary between equivariant line bundles and these functions is not formalised, so the table is settled modulo that dictionary. The groups themselves are known for n even. The quotient $[M_{0,n}/\mathfrak{S}_n]$ is then the stack of projective classes of smooth binary forms of degree n modulo PGL_2, whose Picard group is $\mathbb{Z}/2(n-1)$ when the characteristic does not divide n, by Gorchinskiy and Viviani's Theorem 3.6 [18]. Edidin and Hu, from Pernice's presentation by a Borel subgroup [39], compute the integral Chow ring of $[M_{0,2g+2}/\mathfrak{S}_{2g+1}]$, binary forms with a marked root, as $\mathbb{Z}[l]/(2g(2g+1)\, l)$ for $g \geq 2$, with an explicit generator l [15, Theorem 1.3]; that L_i generates this group follows from the Lean lower bound. For n odd the groups can be computed by the method of Arsie and Vistoli [2]; the edition has recomputed them without finding them stated, and its computation is not checked. What the edition adds is the exponents for n odd and the certification of all the exponents in Lean, in the model.

A refutation outside Lean. Bourbaki's draft no. 307 (1958) on universes [1] asks whether there are universes other than the empty one and the U_n, and answers that it seems probable. Under the first definition given there, with every object in some U_n, the answer is no; under the second, which the text says comes to the same thing, the answer is not decided in ZFC with those axioms, since the two definitions differ at every inaccessible cardinal that is a limit of inaccessibles. The argument, which uses the classification of universes as V_κ under foundation [46], is written out in the edition's register and not formalised: Mathlib has no theory of Grothendieck universes in this form. By contrast, two Bourbaki drafts are formalised: no. 288, on the exponential law for compact convergence, and a refutation of Corollary 6 of Proposition 14 of no. 457 (1965), which equates the degree of an integral separable algebra over a field with its number of points. In infinite degree it fails: the locally constant functions on the Cantor set have countable degree and a continuum of points.

Table 2. Seven conjectures in the fonds, from the edition's note on conjectures. Dates are of the leaves or of the inventory, whose brackets mark an inferred date; the leaf dates are machine readings of the transcriptions, not checked on the facsimile. The last column says whether Mathlib can state the conjecture today.

Conjecture	First in the fonds	Published source	Statable in Mathlib
Standard conjectures	14, p. 44, letter to Serre, 27 Aug. 1965	Grothendieck 1969 [21]; Kleiman 1968, 1994 [31,32]	no Weil cohomology
Period conjecture (de Rham–Betti)	13, pp. 47–48 (1965)	Grothendieck 1966 [19]; Bost–Charles 2016 [8]	elliptic case
Generalised Hodge (ℓ-adic form)	12, pp. 2–3 (1964–[c. 1971])	Grothendieck 1969 [20]	no
p-curvature	93, pp. 193–197 ([1972]–1981)	Katz 1970, 1972 [29,30]	trivial bundle only
Homotopy hypothesis	134-2, p. 42, letter to Breen, 5 Feb. 1975	*Pursuing Stacks* (1983) [36]	partly
Galois action, $\widehat{GT}$	142 ([from 1978]); 119, 147	*Esquisse* (1984, publ. 1997) [23]	partly
Section conjecture	149, p. 64 (undated)	letter to Faltings, 1983 [22]	no étale π_1

6 Conjectures in the Notes

A reading of the whole fonds shows seven well-known conjectures, some earlier than the texts usually cited for them (Table 2). The edition makes no claim of priority: a conjecture on a leaf dates when he wrote it down and says nothing of whether others had.

None of the seven was in Google DeepMind's *Formal Conjectures* repository [17] when the edition searched it on 17 October 2026. That repository collects open conjectures, and some solved variants, stated in Lean with a cited source (the literature, problem lists, Wikipedia, MathOverflow, the OEIS), and the edition's note on conjectures was posted on the Lean community's channel for it. Three of the seven cannot be stated in Mathlib today, because the objects they speak of do not exist there: a Weil cohomology theory, Hodge structures on the cohomology of a variety, the étale fundamental group. A faithful statement needs those definitions first, and one written with ad hoc substitutes would be a different conjecture. The other four can be stated only in part. The edition instead proves the parts around a conjecture that rest on existing theories, and states the conjecture in prose beside them.

Folder 93 shows the method. Pages 193–194 make a remark and ask a question. If a connection defined over the integers has finite monodromy, its reductions modulo p have vanishing p-curvature for almost all p; is the converse true? That converse is the Grothendieck–Katz p-curvature conjecture. Lean proves, over a commutative ring of characteristic p with a derivation D, that D^p is again a derivation, Jacobson's formula for the p-th power of $D + \mu_a$, and the easy direction on the trivial bundle: in rank one, a connection trivialised by a unit has vanishing p-curvature, and in rank n, when $D^p = 0$, so does a connection with a horizontal basis (`curvMat_eq_zero_of_horizontal`), with worked examples in rank two, the Airy connection among them. The appendix then states the conjec-

ture in full, with its known cases, Katz's for Gauss–Manin connections [30], and the solvable case through Bost and others [7], and says that Lean does not touch the general case. Around the Galois action on Teichmüller towers, Lean proves the lemma of page 22 of folder 140-1, that the $\mathfrak{S}_3$-equivariant Picard group of $M_{0,3}$, $H^1(U_{0,3}, \mathfrak{S}_3; \mathbb{G}_m)$, is $\mathbb{Z}/6$ in a model of the units of $U_{0,3}$, and the entries of folder 146 above. Around the standard conjectures, the Kleiman match of folder 16 is formal; the step of the same folder that splits the critical-dimension condition A^0 into two one-sided restriction conditions A'^0 and A''^0 was not found in Kleiman's 1968 paper and stays a candidate.

The notes also pose questions with no printed source known to the edition, such as a description of the graph of the 27 lines on a cubic surface around a marked triangle as a torsor under a Baer product of quaternion extensions (folder 69), or a bound on the cohomological dimension of affine schemes of mixed characteristic, with residue characteristic ℓ, by the degree of imperfection (folder 31). Such questions would need a citable source to enter *Formal Conjectures*. For folder 69 the group-theoretic part is in Lean: the automorphisms preserving a triangle form a group of order 1152, checked through a stabiliser chain. For folder 31 the local form holds in a sharper form on an open of the spectrum of an excellent henselian local domain, by Gabber and Orgogozo [16, Theorem 1.2 and §6], and Mathew bounds the cohomological dimension of the residue fields of $R[1/p]$ for R an excellent normal domain henselian along an ideal containing p [37, Proposition 4.5]. For an arbitrary affine scheme the best general bound we found is the cruder $\dim X + \sup_x \operatorname{cd}_p k(x)$ [13, Corollary 3.29], and the global bound with its « $+\,3$ » was not found in print.

7 Contributing to the Tau Ceti Roadmaps

Tau Ceti [42] is a Lean 4 library downstream of Mathlib that accepts code written by AI systems. People steer it through a separate repository of roadmaps [43]. A roadmap is a `README.md` that specifies one area as an ordered set of definitions and statements, usually with a file `Suggested.lean` of target signatures whose theorems are left as `sorry`. People write and approve the roadmaps, and the proofs are written in the code repository, by provers among others. Axiom's formalisation of the classification of finite simple groups [3] is one such user: its Lake manifest pins Tau Ceti at a fixed revision.[3]

The author has proposed eight roadmaps written from the edition's needs, all still under review on 29 October 2026: the profinite Grothendieck–Teichmüller group (#1058; folders 119, 140-1, 142, 145, 146 and 147), elliptic functions, periods and complex multiplication (#1059; folders 13 and 18), Grothendieck ∞-groupoids, with globular theories, coherators and their models (#1060; folder 134-2), Lefschetz structures and algebraic operators on cohomology (#1061; folders 14, 16 and 50), the homotopy theory of small categories, with test categories, basic localisers, derivators and Picard stacks (#1521; folders 76, 104, 106, 107,

[3] `github.com/AxiomMath/CFSG`, file `lake-manifest.json`, read on 27 October 2026.

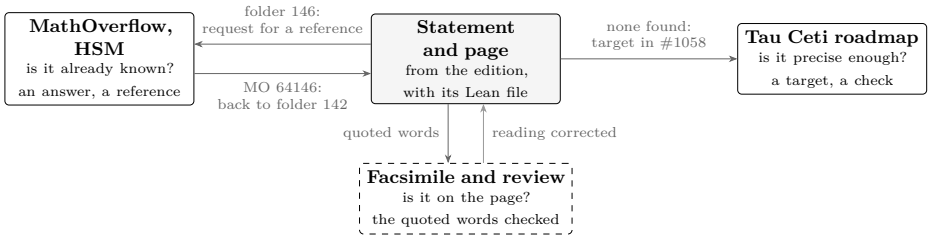


Fig. 3. One statement, three public contributions. A question on MathOverflow or History of Science and Mathematics tests whether the statement is known, a Tau Ceti roadmap whether it is precise enough to formalise, and the facsimile whether it is on the page. The arrows give the two cases of Section 7: the odd case of folder 146, sent out as a request for a reference and entered in #1058; and a 2011 question on the Grothendieck–Teichmüller conjecture, answered from folder 142, whose quoted words went to the review page.

114 and 118), group schemes over a base (#1522; folders 72 and 90), pseudo-line arrangements and oriented matroids (#1523; folders 72, 79, 80, 86 and 154), and equivariant intersection theory and Picard groups of quotient stacks (#1685; folder 146). Each names its folders and pages, and the Lean file and theorem of the edition where one exists, under the rule of Section 3. A ninth, of the general lemmas that the proofs needed and that Mathlib lacks (#1431), specifies the 26 files under `lean/Grothendieck/Upstream/`, so that contributors can write them to the library's conventions.

A roadmap fed by such an edition offers a prover five things that a roadmap written from textbooks does not. The first is *targets*: exact statements, each with its page, that were never printed; #1058, for instance, states as a target the trivialising exponents of folder 146, p. 70, for every n. The second is *proof plans*: a draft gives the order of the lemmas, as folder 16, pp. 47–48, does for the linear algebra of the Lefschetz operator in #1061. The third is *dead ends*: statements that the formalisation refuted (Section 5) enter the roadmap as checks with their counterexamples, such as the classifying-space counterexample recorded under the asphericity of presheaves in #1521 (folder 118, pp. 33–35), so that no search is spent on them. The fourth is *building blocks*: the edition's compiled theorems and the general lemmas of `Upstream/`, which a contributor can restate in Tau Ceti instead of proving again. The fifth is *provenance*: the roadmap says who stated what and where, so that a result proved later from it can be attributed correctly.

The same work makes a triple public contribution (Fig. 3): an answer on a question-and-answer site, a target in a roadmap, and a reading checked on the facsimile. The odd case of folder 146 was put to MathOverflow as a request for a reference;[4] none turned up beyond the even case of Edidin and Hu, and the statement entered #1058 with its Lean proof. In the other direction, an

[4] `https://mathoverflow.net/q/516243`.

unanswered question of 2011 on the evidence for the Grothendieck–Teichmüller conjecture[5] led back to folder 142, pp. 82–103, where the conjecture is stated for curves; answering it showed that the modernised reading quoted the decisive sentence in a reconstructed wording, since corrected, and sent the quoted words to the review page. The loop needs only a statement, its page and a link, so it carries over to other archives.

These roadmaps reach the layers underneath Grothendieck's large conjectures, not the conjectures themselves. The standard conjectures, the anabelian conjectures and the section conjecture need étale cohomology, topoi and motives, which neither Mathlib nor Tau Ceti has yet, and the four roadmaps specify foundations, examples and counterexamples on the way there. The same work is planned for the unpublished drafts of the Bourbaki archive and for Quillen's notebooks (issue #95). Only statements and page references go upstream, never transcriptions, whose rights belong to others.

8 Human Checking and Its Limits

The formalisation checks the readings; it checks neither the transcriptions nor the link between them. The transcriptions are spot-checked. A page of the site, `/review/`, turns the edition's checklist into single questions, each on one spot of one page, shown beside Montpellier's facsimile with a fixed set of answers; answers come back as issues in the repository and are applied at source, with a dated revision line. The register of answers (`src/content/review-answered.json`) records 183 items answered through 19 issues. In 167 the answer was applied or confirmed the entry as it stood; the others found the spot still illegible or left it open. The settled items include 45 dates checked on their leaves and 40 insertions whose hand was in question. Other points the author has checked against the facsimile are recorded, dated, in the batch files they concern. No batch has been read whole by a person.

One answer set shows the limit. In issue #69 the reader, who reads the hand, answered « can't tell » to the items that asked whether one of the edition's own arguments holds or whether a reference cited from memory is right, and most of the confirmations given earlier on such items were withdrawn. Reading the hand does not equip a reader to judge the mathematics, and a Lean proof covers part of that need for the statements it formalises. For the faithfulness of a statement to its page, and for everything in tier 3, the edition has no check but a specialist's, and labels its machine second readings as such.

9 Costs and Reproducibility

The repository records costs in shares of a weekly usage allowance of the author's subscription, not in tokens or currency, and only for some work. A second transcription pass costs about 1% of a week per twenty-page batch (issue #51),

[5] `https://mathoverflow.net/q/64146`.

and twenty-one batches of Quillen's notebooks were estimated at about 10% (#42). For the Lean work no cost per proof is recorded. Its scale is visible in the history: 206 commits touching `lean/` from 26 September 2026 to 24 October 2026, issue estimates of 80 to more than 1,200 lines per finding (#54, #72), and rows dropped when the estimate grew beyond their value. The constraint in practice was the weekly allowance, which bounded how many proofs ran in parallel. The code and the site are released under CC0. At commit `81dcfaf` the Lean files build with `cd lean && lake exe cache get && lake build`, which fetches the pinned Mathlib; the axiom audit runs in CI, and `#print axioms` gives the axioms of any one theorem.

10 Related Work

Mathlib [44] on Lean 4 [38] is the library everything here depends on. Large formalisations led by mathematicians, the Kepler conjecture [24], the Liquid Tensor Experiment [40,10] and the Fermat's Last Theorem project [9,6], formalise a known proof towards a stated theorem. This edition goes the other way: it starts from statements scattered through drafts and asks which of them can be checked at all. Formal Abstracts [25] proposed to state the main results of published papers formally without their proofs; Formal Conjectures [17] does this for open problems with a cited source. The statements here have no published source of their own, which is why the attribution rule of Section 3 is needed.

Old mathematical texts have been formalised before. Beeson, Narboux and Wiedijk checked the propositions of Book I of Euclid's *Elements* by computer [5]; there, as here, the formal statement is a reading of the text, but Euclid's statements are fixed and published, whereas these are found in drafts by a machine reading. Kohlhase's sTEX uses LATEX macros as semantic markup for mathematics [33]; the restricted LATEX of the transcriptions uses macros for editorial semantics (illegible, struck, supplied) and leaves the mathematics unannotated, so that the formal statement is the first place where its meaning is fixed.

Autoformalisation was set as a goal by Szegedy [41]. Wu et al. [47] showed that large language models can translate competition problems into formal statements, ProofNet [4] gave a benchmark at undergraduate level, and DeepSeek-Prover [48] trained provers on synthetic formal data. These works measure translation and proof on curated statements. Here the statements come from manuscripts read by a model, so a failure can lie in the transcription, the reading, the formal statement or the mathematics, and only the last two are within Lean's reach.

Mathematicians' Nachlässe have been edited by hand, as the *Correspondance Grothendieck–Serre* [14], *Les Dérivateurs* [34] and *À la poursuite des champs* [36] were. The edition's transcription method and its encoding in TEI are reported separately [28].

11 Limits and Future Work

Most of the open statements are out of reach until Mathlib has the theories they speak of, and the edition will not replace them by special cases of its own choosing. While the reading views of the transcriptions are withheld, a reader can check a theorem against the facsimile only through the statement and page reference that each card gives. The 902 drawings that the transcriptions describe in prose (issue #73) are not yet encoded, and some statements depend on them. The tooling, built for this fonds and reused for other archives, among them the Bourbaki archive and Quillen's notebooks, could become a generic tool for editing mathematical manuscripts into a checked edition (issue #79). The checks still missing are a specialist's reading of each formal statement against its page, and of the edition's own arguments.

Use of Generative AI

This work used language models of Anthropic's Claude family, through Claude Code, at every step marked *model* in Fig. 1. The first transcription passes of the 638 batches were written by Claude Opus 5.5 (505 batches), Claude Opus 5 (102), Claude Fable 5.1 (26) and Claude Fable 5 (5), as each batch file records. The modernised readings and the findings were written by the same four models. Claude Opus 5.5 wrote the Lean proofs, the appendix, the scripts, the pull requests to Mathlib and the roadmaps proposed to Tau Ceti, with Claude Sonnet 5.5 in one commit, the move of the Mathlib pin, and drafted this paper from the repository's files and history. The author chose the folders, and the statements to formalise among those the models proposed, set the editorial and attribution rules, checked readings against the facsimile where the paper says so, revised the paper and decided what to publish, and takes responsibility for its content. Lean checks the proofs mechanically; the transcriptions have not been checked against the facsimiles in full by a person, and the paper says so wherever a claim depends on them.

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References

1. Archives Henri Poincaré: Archives de l'Association des Collaborateurs de Nicolas Bourbaki. `https://archives-bourbaki.ahp-numerique.fr/` (2026)
2. Arsie, A., Vistoli, A.: Stacks of cyclic covers of projective spaces. Compositio Mathematica **140**(3), 647–666 (2004). `https://doi.org/10.1112/S0010437X03000253`
3. Axiom: The classification of finite simple groups, formalized in Lean. `https://cfsg.axiommath.ai/` (2026), accessed 28 October 2026
4. Azerbayev, Z., Piotrowski, B., Schoelkopf, H., Ayers, E.W., Radev, D., et al.: ProofNet: Autoformalizing and formally proving undergraduate-level mathematics. arXiv:2302.12433 (2023)

5. Beeson, M., Narboux, J., Wiedijk, F.: Proof-checking Euclid. Annals of Mathematics and Artificial Intelligence **85**(2–4), 213–257 (2019). https://doi.org/10.1007/s10472-018-9606-x
6. Best, A.J., Birkbeck, C., Brasca, R., Rodriguez Boidi, E.: Fermat's Last Theorem for regular primes. arXiv:2305.08955 (2023)
7. Bost, J.B.: Algebraic leaves of algebraic foliations over number fields. Publications Mathématiques de l'IHÉS **93**, 161–221 (2001). https://doi.org/10.1007/s10240-001-8191-3
8. Bost, J.B., Charles, F.: Some remarks concerning the Grothendieck period conjecture. Journal für die reine und angewandte Mathematik **714**, 175–208 (2016). https://doi.org/10.1515/crelle-2014-0025
9. Buzzard, K., et al.: The Fermat's Last Theorem project. https://imperialcollegelondon.github.io/FLT/, accessed 21 October 2026
10. Castelvecchi, D.: Mathematicians welcome computer-assisted proof in 'grand unification' theory. Nature **595**, 18–19 (2021). https://doi.org/10.1038/d41586-021-01627-2
11. Cisinski, D.C.: Les préfaisceaux comme modèles des types d'homotopie. No. 308 in Astérisque, Société mathématique de France, Paris (2006). https://doi.org/10.24033/ast.715
12. Cisinski, D.C.: Answer to MathOverflow question 516223. https://mathoverflow.net/questions/516223 (2026), 21 October 2026
13. Clausen, D., Mathew, A.: Hyperdescent and étale K-theory. Inventiones mathematicae **225**(3), 981–1076 (2021). https://doi.org/10.1007/s00222-021-01043-3
14. Colmez, P., Serre, J.P. (eds.): Correspondance Grothendieck–Serre. No. 2 in Documents mathématiques, Société mathématique de France, Paris (2001)
15. Edidin, D., Hu, Z.: The integral Chow rings of the stacks of hyperelliptic Weierstrass points. Michigan Mathematical Journal (2025). https://doi.org/10.1307/mmj/20236479, arXiv:2208.00556
16. Gabber, O., Orgogozo, F.: Sur la p-dimension des corps. Inventiones Mathematicae **174**(1), 47–80 (2008). https://doi.org/10.1007/s00222-008-0133-y
17. Google DeepMind: Formal conjectures. https://github.com/google-deepmind/formal-conjectures, accessed 21 October 2026
18. Gorchinskiy, S., Viviani, F.: Picard group of moduli of hyperelliptic curves. Mathematische Zeitschrift **258**, 319–331 (2008). https://doi.org/10.1007/s00209-007-0173-9
19. Grothendieck, A.: On the de Rham cohomology of algebraic varieties. Publications Mathématiques de l'IHÉS **29**, 95–103 (1966). https://doi.org/10.1007/BF02684807
20. Grothendieck, A.: Hodge's general conjecture is false for trivial reasons. Topology **8**(3), 299–303 (1969). https://doi.org/10.1016/0040-9383(69)90016-0
21. Grothendieck, A.: Standard conjectures on algebraic cycles. In: Algebraic Geometry (Internat. Colloq., Tata Inst. Fund. Res., Bombay, 1968), pp. 193–199. Oxford University Press, London (1969)
22. Grothendieck, A.: Brief an G. Faltings. In: Schneps, L., Lochak, P. (eds.) Geometric Galois Actions 1, pp. 49–58. No. 242 in London Mathematical Society Lecture Note Series, Cambridge University Press (1997). https://doi.org/10.1017/CBO9780511758874.004
23. Grothendieck, A.: Esquisse d'un programme. In: Schneps, L., Lochak, P. (eds.) Geometric Galois Actions 1, pp. 7–48. No. 242 in London Mathematical Society Lecture Note Series, Cambridge University Press (1997). https://doi.org/10.1017/CBO9780511758874.003

24. Hales, T., Adams, M., Bauer, G., Dang, T.D., Harrison, J., Hoang, L.T., et al.: A formal proof of the Kepler conjecture. Forum of Mathematics, Pi **5** (2017). https://doi.org/10.1017/fmp.2017.1
25. Hales, T., et al.: Formal abstracts. https://formalabstracts.github.io/, accessed 21 October 2026
26. Hua, M.: Annexe. Énoncés des lectures modernisées démontrés. Lean appendix of the edition, https://grothendieck.commutator.io/article/grothendieck-lean.pdf (2026)
27. Hua, M.: grothendieck-project: an edition of the Grothendieck working notes. https://github.com/Commutator-IO/grothendieck-project (2026), commit 8aecaa3, 24 October 2026
28. Hua, M.: A machine reads Grothendieck's working notes: Transcribing and encoding mathematical manuscripts. Draft prepared for the Journal of the Text Encoding Initiative, https://grothendieck.commutator.io/article/grothendieck-jtei.pdf (2026)
29. Katz, N.M.: Nilpotent connections and the monodromy theorem: Applications of a result of Turrittin. Publications Mathématiques de l'IHÉS **39**, 175–232 (1970). https://doi.org/10.1007/BF02684688
30. Katz, N.M.: Algebraic solutions of differential equations (p-curvature and the Hodge filtration). Inventiones Mathematicae **18**, 1–118 (1972). https://doi.org/10.1007/BF01389714
31. Kleiman, S.L.: Algebraic cycles and the Weil conjectures. In: Dix exposés sur la cohomologie des schémas, Advanced Studies in Pure Mathematics, vol. 3, pp. 359–386. North-Holland, Amsterdam (1968), zbMATH 0198.25902
32. Kleiman, S.L.: The standard conjectures. In: Motives, Proceedings of Symposia in Pure Mathematics, vol. 55, Part 1, pp. 3–20. American Mathematical Society (1994). https://doi.org/10.1090/pspum/055.1/1265519
33. Kohlhase, M.: Using LaTeX as a semantic markup format. Mathematics in Computer Science **2**(2), 279–304 (2008). https://doi.org/10.1007/s11786-008-0055-5
34. Künzer, M., Malgoire, J., Maltsiniotis, G.: Alexandre Grothendieck, les dérivateurs (online edition). https://webusers.imj-prg.fr/~georges.maltsiniotis/groth/Derivateurs.html
35. Maltsiniotis, G.: La théorie de l'homotopie de Grothendieck. No. 301 in Astérisque, Société mathématique de France, Paris (2005). https://doi.org/10.24033/ast.689
36. Maltsiniotis, G. (ed.): Alexandre Grothendieck, À la poursuite des champs, vol. I. No. 20 in Documents mathématiques, Société mathématique de France, Paris (2022)
37. Mathew, A.: On K(1)-local TR. Compositio Mathematica **157**(5), 1079–1119 (2021). https://doi.org/10.1112/S0010437X21007144
38. de Moura, L., Ullrich, S.: The Lean 4 theorem prover and programming language. In: Automated Deduction – CADE 28. Lecture Notes in Computer Science, vol. 12699, pp. 625–635. Springer (2021). https://doi.org/10.1007/978-3-030-79876-5_37
39. Pernice, M.: The integral Chow ring of the stack of 1-pointed hyperelliptic curves. International Mathematics Research Notices **2022**(15), 11539–11574 (2022). https://doi.org/10.1093/imrn/rnab072
40. Scholze, P.: Liquid tensor experiment. Experimental Mathematics **31**, 349–354 (2022). https://doi.org/10.1080/10586458.2021.1926016

41. Szegedy, C.: A promising path towards autoformalization and general artificial intelligence. In: Intelligent Computer Mathematics (CICM 2020). Lecture Notes in Computer Science, vol. 12236, pp. 3–20. Springer (2020). https://doi.org/10.1007/978-3-030-53518-6_1
42. Tau Ceti Project: Tau ceti: an AIs-welcome Lean 4 library downstream of Mathlib. https://github.com/TauCetiProject/TauCeti, accessed 28 October 2026
43. Tau Ceti Project: Tau ceti roadmap: human-controlled roadmaps for Tau Ceti. https://github.com/TauCetiProject/TauCetiRoadmap, accessed 28 October 2026
44. The mathlib Community: The Lean mathematical library. In: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020). pp. 367–381. ACM (2020). https://doi.org/10.1145/3372885.3373824
45. Université de Montpellier: Archives Grothendieck. https://grothendieck.umontpellier.fr/
46. Williams, N.H.: On Grothendieck universes. Compositio Mathematica **21**, 1–3 (1969)
47. Wu, Y., Jiang, A.Q., Li, W., Rabe, M.N., Staats, C., Jamnik, M., Szegedy, C.: Autoformalization with large language models. In: Advances in Neural Information Processing Systems 35 (NeurIPS 2022). pp. 32353–32368 (2022). https://doi.org/10.52202/068431-2344
48. Xin, H., Guo, D., Shao, Z., Ren, Z., Zhu, Q., et al.: DeepSeek-Prover: Advancing theorem proving in LLMs through large-scale synthetic data. arXiv:2405.14333 (2024)